Part 2

Question 1: How Websites Remember You (State Preservation)

Websites on the internet are naturally forgetful—every time you click a new page, the server treats it as a brand-new visit and has no memory of who you are. To fix this so you don't have to log in on every single click, websites use cookies and sessions.

When you log into an account, the website checks your password and creates a temporary "session" file on its server. It then gives your browser a small digital ticket called a cookie. Every time you click a new link on that site, your browser automatically hands that cookie back to the server like a VIP pass. The server reads the pass, checks its records, and says, "Ah, this is John!" This trick allows the website to keep you logged in, save items in your online shopping cart, or remember your preferences as you move around. Without cookies and sessions, the internet would be exhausting because every single page click would log you out and wipe your data clean.

Question 2: Moving Django Data to MariaDB

By default, Django projects store information in a lightweight, built-in file database called SQLite. When you are ready to launch your application for real-world use, you often need to upgrade to a more powerful server-based database like MariaDB.

To make this switch, you first need to install a helper tool (a database driver) so Python can talk to MariaDB. Next, you open your Django settings file and update your database configuration. Instead of pointing to SQLite, you change the engine setting to MySQL and fill in your MariaDB server details, such as the database name, username, password, and port number.

Once your project knows where to look, you run two simple terminal commands. First, you run a command to prepare your data blueprints. Second, you run the migration command, which tells Django to connect to your MariaDB server and automatically build all the required tables, rows, and columns. If you have any older test data you want to bring over, you can export it and import it into your brand-new MariaDB system.